The Digital WatchDog for Transparent Procurement: Detecting Bid Rigging in China's Government and Enterprise Procurement via Blockchain and Large Language Models

ARTICLE INFO

Keywords:
Bid Rigging
Large Language Models (LLMs)
Blockchain
RegTech

ABSTRACT

Detecting bid rigging remains a costly challenge due to the limitations of traditional price-based screening. This paper proposes a Blockchain + Large Language Models (LLMs) RegTech framework to bridge the gap between structured hard information and unstructured soft information. Using real-world data from a municipal bidding alliance chain in a major southern China's city, we present three key findings. First, the integrated framework breaks the theoretical ceiling of single-source screening by leveraging LLMs to capture textual risk signals (e.g., semantic similarities) that statistical models miss. Second, we identify a counter-intuitive Less is More phenomenon where simple Zero-shot strategies outperform complex Chain-of-Thought prompting, reducing signal noise in high-dimensional fraud detection. Third, counterfactual experiments validate Blockchain as a ~~non-negotiable~~crucial Trust Anchor, without which LLMs detection ~~degenerates into random guessing~~performance significantly degrades under insider tampering. We suggest deploying this low-cost, dual-drive mechanism for real-time procurement supervision.

1. Introduction

Bid rigging severely undermines market competition and the integrity of Transparent Procurement. However, existing regulatory mechanisms face significant efficiency bottlenecks. Traditional audits and statistical screens (Imhof, Karagök and Rutz, 2018), even when enhanced by machine learning (Huber, Imhof and Ishii, 2022) or economic models (Kawai and Nakabayashi, 2022; Chassang, Kawai, Nakabayashi and Ortner, 2022), face two fundamental limitations. First, they prioritize numerical hard information while systematically ignoring unstructured soft information in bid documents (Liberti and Petersen, 2019). Despite evidence that textual data holds critical risk signals (Hu, Li and Liu, 2022) and LLMs proven potential in finance (Dowling and Lucey, 2023), their application in forensic procurement auditing remains underexplored.Crucially, while previous text-based screening methods (e.g., TF-IDF, keyword matching) rely on static lexical matching within bag-of-words frameworks, our LLM-based approach performs cross-modal contextual reasoning. This enables the detection of semantically sophisticated collusion patterns where technical details may differ superficially but reveal logical inconsistencies and deeper disguises through contextual understanding. Second, manual verification is costly and lacks immutable audit trails, allowing evasion and eroding public trust.

To bridge this RegTech gap, we propose a Digital Watchdog framework integrating Blockchain and Large Language Models (LLMs), utilizing real-world data from a government bidding alliance chain. First, regarding infrastructure, Blockchain serves as a ~~non-negotiable~~crucial Trust Anchor rather than mere storage. Our counterfactual experiments confirm that without cryptographic finality to prevent insider tampering, even advanced algorithms degenerate into random guessing (Garbage In, Garbage Out) (Cong and He, 2019). Second, regarding detection, we deploy LLMs to overcome traditional models dependency on hard information. Unlike econometric models, LLMs interpret unstructured soft information (e.g., semantic similarities), thereby filling regulatory blind spots. Finally, we evaluate Zero-shot, Few-shot, and Chain-of-Thought (CoT) prompting strategies to optimize the signal-to-noise ratio for automated supervision.

This paper makes three contributions. First, we quantify the incremental value of Soft Information in detecting bid rigging. Complementing macro-level fraud studies (Li and Pan, 2023), we focus on micro-level information aggregation. We demonstrate that integrating unstructured textual risks enables the framework to break the theoretical ceiling of single-source numerical screening. Second, we validate Blockchain as an essential Trust Anchor for AI governance. Unlike studies viewing blockchain merely as storage (Nasir, Shaukat and Hameed, 2021), we prove via tampering simulations that cryptographic immutability is a prerequisite for effective supervision against insider threats. Third, we

ORCID(s):

identify a counter-intuitive Less is More phenomenon where simple Zero-shot strategies outperform complex reasoning chains. We attribute this to the signal-to-noise ratio: excessive context tends to introduce noise, causing the model to overfit irrelevant details while ignoring salient risk indicators.

2. Method and Data

2.1. Governance Framework Based on Blockchain

The monitoring mechanism proposed in this paper operates within an alliance chain environment based on Blockchain technology. Blockchain is not merely a database but a commitment mechanism capable of mitigating data trust issues in procurement audits. By recording the hash values of all bidding documents, it ensures that the information set available to regulators at a certain moment cannot be tampered with. This approach leverages the transparency and information disclosure benefits of blockchain technology widely validated in supply chain management literature (Chod, Trichakis, Tsoukalas et al., 2020; Cao, Gunasekaran and Dubey, 2023). This characteristic is crucial for the empirical strategy, as it isolates ex-ante collusion (bid rigging) from ex-post corruption (bid tampering), ensuring that any anomalies detected by LLMs originate from the bidders collusive behavior rather than internal data manipulation.

2.2. Dataset

This study utilizes a real desensitized dataset obtained from a municipal bidding alliance chain in a southern China's city. This blockchain infrastructure is government-backed and widely deployed for public procurement management. The desensitization process strictly adheres to the Measures for the Administration of Government Procurement Data Security.To prevent label-feature endogeneity (i.e., the model memorizing a blacklist of colluding entities), all specific provider names, employee names, and contact information fed into the model were strictly anonymized or hashed (e.g., replaced with Provider_A, Employee_B). This ensures that the LLMs detection relies solely on behavioral patterns rather than entity identifiers.

The dataset comprises over 200 procurement projects and over 1000 related bidding records.The unit of analysis in this study is the Bidder-Project Pair, where each observation represents a unique combination of a bidder participating in a specific procurement project. In terms of sample composition, it includes 100+ collusive bidding records (approx. 10%) and 800+ competitive bidding records (approx. 90%). This distribution aligns with the collusion occurrence rates observed in real-world procurement markets, ensuring the samples representativeness. ~~The classification is grounded in historical regulatory records and conclusive audit evidence (e.g., physical MAC address collisions and abnormally identical contact information)~~.The ground truth labels (is_collusion = 1) are strictly derived from finalized administrative penalty decisions with conclusive forensic evidence (e.g., physical MAC address collisions indicating shared device usage, identical metadata in bid documents). Ambiguous or disputed cases were excluded from the sample to ensure label reliability. The dataset includes multidimensional features:

- Project Identification: Unique identifier (bid_ann_guid) and tender notice ID.
- Bidder Characteristics: Provider name (x_providername), bidder name (x_bidder), and encrypted contact information.
- Core Bidding Data: Bid price (x_price), price reduction rate (x_pricedecreaserate), duration, and winning status.
- System Trace: Creator/Modifier IDs and blockchain timestamps (created_time, sync_time).

2.3. Identification and Detection Mechanism Based on LLMs

This paper deploys lightweight large language models GLM-4-Flash and GPT-4o-mini as the core detection engines. LLMs directly process raw unstructured bidding data. Consistent with Hansen and Kazinnik (2023), who demonstrated LLMs' proficiency in deciphering nuanced official texts, we leverage this semantic capability to detect subtle collusion signals. In addition to price data, model inputs also include bidder identity, historical interaction graphs, and text remark information. To evaluate the adaptability of the models in bidding supervision tasks, this paper tests three prompting strategies:

- Zero-shot: Only provides the model with raw data and direct instructions (e.g., Identify potential collusion based on price dispersion and bidder associations), without providing any prior examples.

- Few-shot: Provides three examples of collusive bidding and competitive bidding before asking the model to classify the target project.
- Chain-of-Thought (CoT): Instructs the model to explicitly show reasoning steps before reaching a conclusion (e.g., Step 1: Calculate price variance; Step 2: Check shared contact information, etc.).

2.4. Benchmark Setting

To quantify the incremental contribution of the LLMs method, this paper ~~sets a traditional statistical benchmark widely used in antitrust screening: the Coefficient of Variation Screen (CV Screen). Although sophisticated screening methods based on machine learning and order statistics have been proposed in recent high-level studies (Huber et al., 2022; Huber the classic CV Screen remains the standard baseline for capturing the economic intuition that collusive bids exhibit unnaturally low variance (Imhof et al., 2018). When the coefficient of variation of bid prices is lower than a specific threshold (e.g., CV < 0.05), the project is flagged as suspicious.~~establishes a comprehensive benchmark suite comprising three progressively sophisticated models: (1) Logistic Regression, serving as the interpretable linear baseline that captures fundamental statistical relationships in bid data; (2) LightGBM (Ke et al., 2017), a gradient boosting framework optimized for efficiency and handling categorical features; and (3) XGBoost (Chen and Guestrin, 2016), a regularized boosting algorithm widely adopted in fraud detection applications. This multi-model benchmark allows us to isolate the incremental value of LLM-based soft information processing beyond what traditional machine learning can achieve with hard information alone.

To ensure reproducibility and fair comparison, all models were evaluated under a unified experimental protocol. The dataset was partitioned into 60% training, 20% validation, and 20% test sets using a fixed random seed (Seed = 42). To assess the robustness of our findings, we conducted sensitivity analyses across 10 different random seeds, with results reported as mean ś standard deviation. The classification threshold for each model was determined on the 20% validation set with the objective of maximizing recall, reflecting the regulatory priority of minimizing false negatives (i.e., failing to detect actual collusion) in procurement supervision contexts. The subsequent text will compare the performance of the LLM method and the traditional ~~econometric~~machine learning benchmarks in terms of Precision, Recall, and F1 Score.

In bid rigging detection, the costs of classification errors are highly asymmetric. A false negative (missing an actual collusion case) undermines market competition and erodes public trust, whereas a false positive merely triggers additional manual review without irreversible harm. This reflects the regulatory principle of rather flag more for review than let collusion escape. Consequently, the LLM framework serves as a high-sensitivity screening tool to identify all suspicious projects for expert inspection, rather than rendering final judgments. Under this paradigm, Recallthe proportion of actual collusion cases correctly identifiedbecomes the most critical metric. Therefore, the classification threshold was calibrated on the validation set with the objective of maximizing Recall.

3. Empirical Results

3.1. Analysis of Detection Performance Based on LLMs

To rigorously evaluate the effectiveness of the proposed framework, we first ~~established a traditional econometric benchmark using a Logit Regression Model. The Logit Regression Model was trained solely on structured "hard information" (e.g., price variance, price difference rates, and bidder counts). The empirical results indicate that the Logit model achieves a detection accuracy of 0.6869, effectively capturing the linear relationships in numerical bid data.~~a comprehensive machine learning benchmark suite. All baseline modelsLogistic Regression, LightGBM, and XGBoostwere trained exclusively on structured hard information (e.g., price variance, price difference rates, and bidder counts). Table 1 presents the comparative performance of these benchmarks against our LLM-based framework on the held-out test set.

The results in 1 reveal several important findings. First, among the traditional ML benchmarks, XGBoost achieves the highest F1-Score (0.800), followed by LightGBM (0.788) and Logistic Regression (0.750). However, all three models exhibit a common limitation: while maintaining high precision (0.9331.000), their recall rates remain relatively low (0.6000.700), indicating a tendency to miss actual collusion cases. Second, the LLM-based framework (GLM-4-Flash, Zero-shot) outperforms all traditional benchmarks across key metrics, achieving the highest ROC-AUC (), Recall (), and F1-Score (). This superior recall is particularly valuable in regulatory contexts where failing to detect collusion carries significant social costs.

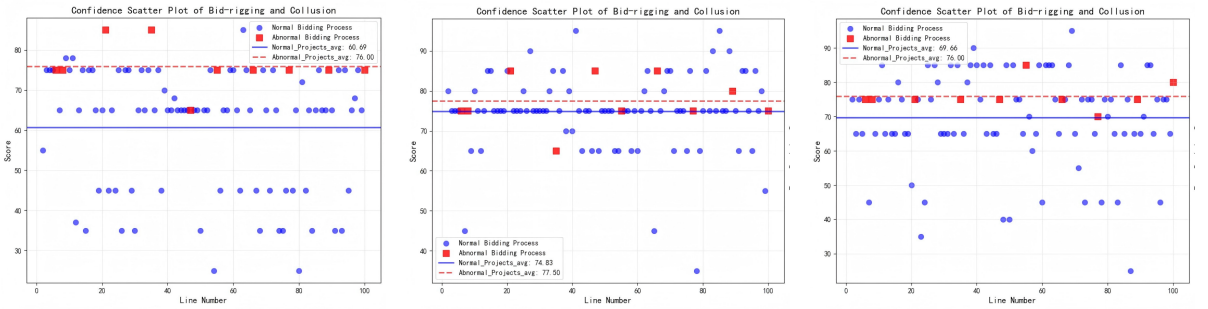
Table 1
Comparative Performance of Detection Models on Test Set

Model	ROC-AUC	PR-AUC	Precision	Recall	F1-Score
Panel A: Traditional ML Benchmarks					
Logistic Regression	0.854	0.737	1.000	0.600	0.750
LightGBM	0.878	0.760	1.000	0.650	0.788
XGBoost	0.854	0.780	0.933	0.700	0.800
Panel B: LLM Framework					
GLM-4-Flash (Zero-shot)					
GPT-4o-mini (Zero-shot)					

Notes: Panel A reports results for traditional machine learning models trained on structured numerical features. Panel B reports results for LLM-based detection incorporating both hard and soft information. Precision, Recall, and F1-Score are reported at the optimal threshold determined on the validation set by maximizing recall. Bold values indicate the best performance in each metric.

Subsequently, we tested the performance of the "Blockchain + LLMs" framework using the GLM-4-Flash base model. Figure 1 displays the confidence score distributions under three experimental settings: Zero-shot, Few-shot, and Chain-of-Thought (CoT).Specifically, the confidence score is a probabilistic assessment metric (ranging from 0 to 100) generated by the LLMs based on the input raw bidding data. Acting as an auditor, the LLM comprehensively analyzes risk signals, integrating unstructured soft information (bid document text similarity) with metadata features; a higher score implies that the LLMs consider the project to have a higher risk of bid rigging. The results reveal two counter-intuitive phenomena:

- **LLMs underperform on pure hard information:** When fed only the same structured numerical data, the optimal accuracy of the LLM (under the Zero-shot strategy) is 0.5455, which is significantly lower than the Logit benchmark (0.6869). This suggests that for low-dimensional numerical tasks, traditional statistical models remain superior to generative LLMs.
- **"Less is More" in Prompting:** As shown in Figure 1, the simple Zero-shot strategy achieves greater differentiation between abnormal and normal bidding projects. In contrast, complex strategies like Few-shot show a minute safety margin (average score difference < 3).



(a) Zero-shot (b) Few-shot (c) Chain-of-Thought

Figure 1: Confidence Score Distributions by Prompting Strategy

Notes: The figure compares the probability density of risk scores for collusive versus competitive bids (Total Sample $N > 500$). The distributional divergence between bid rigging (red) and competitive bidding (blue) is statistically most pronounced in the Zero-shot setting, indicating a higher signal-to-noise ratio.

These findings provide two insights into the theoretical boundaries of LLMs in financial supervision. Extending the discourse on digital technologies and corporate fraud (Yue, Korkmaz and Zhou, 2023), our results suggest that effective supervision requires a hybrid approach.

First, the fact that LLMs lag behind the Logit benchmark on structured data ($0.5455 < 0.6869$) highlights a fundamental ontological distinction: LLMs are probabilistic semantic engines, not deterministic calculation engines. While traditional econometric models isolate linear relationships in numerical variance with mathematical precision, generative AI treats numbers as tokens, lacking the inherent sensitivity to statistical thresholds (e.g., Coefficient of Variation). This ~~proves~~suggests that LLMs cannot simply replace traditional statistical screenings in processing hard information.

Second, the unexpected failure of complex prompting strategies stems from the signal-to-noise ratio problem. In fraud detection tasks involving high-dimensional features, mandating rigid reasoning steps (CoT) or providing specific examples (Few-shot) may induce the model to "hallucinate" logical connections where none exist, or to overfit to irrelevant contextual noise. Consequently, the model becomes distracted from the most salient indicators like price parallelism. Therefore, in automated governance tasks, direct and explicit instructions (Zero-shot) appear more robust than elaborate reasoning chains, enforcing a "Less is More" principle.

3.2. Theoretical Derivation: The Power of Soft Information

Given that LLMs underperform Logit models on structured data, what drives the superior performance of the integrated framework? The answer lies in the incremental value of Soft Information.

Unlike the Logit model which relies solely on structured numerical data (A), our framework integrates unstructured soft information (B). Theoretically, the total identification rate R is not a simple summation but a union probability that accounts for the information overlap. We model the dependency between price anomalies (Hard Information) and textual similarities (Soft Information) using the correlation coefficient ρ. By strictly incorporating the covariance term $\rho\sqrt{Var(A)Var(B)}$ into the joint probability calculation, the system's total efficacy is derived as:

$$R = P(A) + P(B) - P(A \cap B) \tag{1}$$

where the intersection term $P(A \cap B)$ is a function of ρ and the individual variances. This formulation mathematically captures how the dual-channel mechanism fills the regulatory blind spots of single-channel screening.

Empirically, considering the LLM's native advantage in text processing (calculated as $P(B)$) and the observed positive correlation between bid rigging behaviors (ρ), we substitute these derived parameters into the formula. As detailed in **Appendix**, the final calculation yields a total accuracy of: $R = 0.812 > 0.687$ (Logit Benchmark)

This ~~proves~~demonstrates that by effectively aggregating complementary risk signals from unstructured bid documents, the "Blockchain + LLMs" framework breaks the theoretical ceiling of traditional econometric models.

3.3. ~~The Necessity of Blockchain Data Immutability~~The Critical Role of Blockchain Data

A fundamental premise of this study is that the efficacy of any algorithmic detection system is strictly bounded by the integrity of its input data. Effective bid rigging detection depends not merely on the intelligence of the model, but critically on the immutability of the underlying data. To empirically validate the role of Blockchain as ~~an indispensable~~a crucial "trust anchor," this paper conducts a counterfactual experiment simulating a "God's view" tampering scenario. We assume a centralized database environment where a malicious administrator (insider threat) modifies the bid prices of collusive projects ex-post to mimic the statistical characteristics of competitive behavior. Specifically, we employed a "Price Mean Alignment" strategy, where the bid prices of collusive projects were adjusted to fall within the 95% confidence interval of the normal project distribution. This operation was executed by a third party with database administrator privileges, simulating a realistic "insider threat" scenario.

The impact of such tampering on statistical detectability is illustrated in Figure 2. In the original dataset (Left Panel), collusive bids typically exhibit specific distributional anomalies, such as excessive variance or unnatural clustering. However, after the simulated tampering (Right Panel), the distribution of the "cleansed" collusive data becomes statistically indistinguishable from that of genuine competitive bids. The distinct peaks of risk features are smoothed out, effectively bypassing traditional statistical screening mechanisms (such as Benford's Law or variance analysis).

Subsequently, this camouflaged dataset was fed into the LLM detection framework. The results, as presented in Figure 3, reveal a catastrophic degradation in performance. Regardless of the prompting strategy employed (Zero-shot, Few-shot, or CoT), the detection accuracy collapses comprehensively. The AUC values drop to approximately 0.5, indicating that the sophisticated Large Language Model degenerates into random guessing.

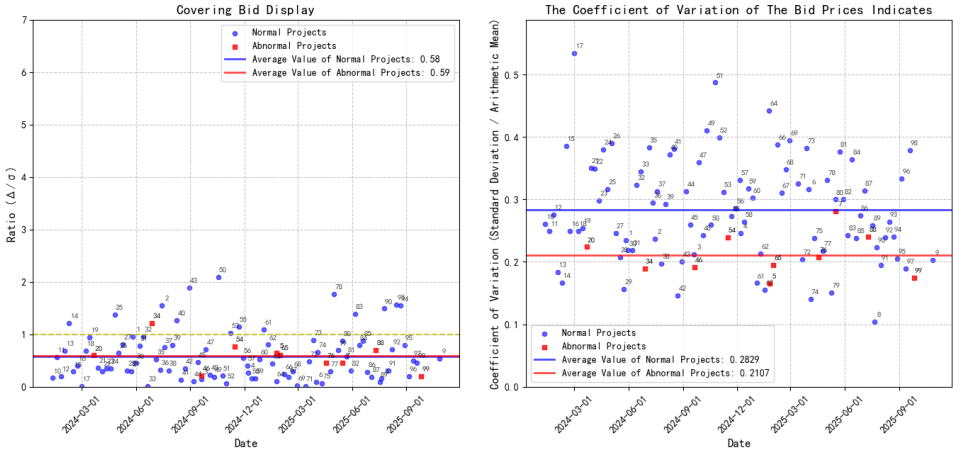


Figure 2: Distribution Changes in Statistical Characteristics of Bid Prices Before and After Data Tampering (Ratio changed to Relative Distance)

This finding leads to a critical conclusion regarding the design of RegTech systems: ~~LLMs without Blockchain is a castle built on sand.~~Without a tamper-proof data infrastructure, even the most advanced LLMs are vulnerable to data manipulation. Even the most advanced LLMs are powerless against purposeful internal tampering. In a centralized system, regulators face the "Garbage In, Garbage Out" dilemma; however, in a blockchain-based system, the distributed ledger locks the information set at the precise moment of submission. This cryptographic finality ensures that the data fed into the AI is the "single version of truth," thereby ~~validating~~suggesting that Blockchain technology is not merely a technical supplement for data storage, but a ~~necessary~~critical institutional infrastructure for automated governance.

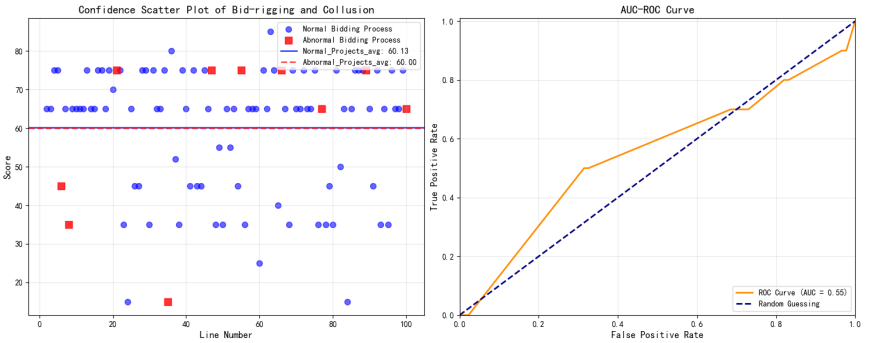
We acknowledge that immutable audit logs with cryptographic verification (e.g., append-only databases with hash chains) represent a viable alternative for ensuring data integrity in certain contexts. However, in multi-stakeholder procurement allianceswhere government agencies, state-owned enterprises, and private bidders operate under mutual distrustsuch centralized solutions face a fundamental limitation: they cannot prevent a single database administrator with root privileges from tampering with records. Blockchains decentralized consensus mechanism addresses this insider threat by distributing trust across multiple independent nodes. No single party, regardless of their administrative privileges, can unilaterally alter historical records without detection by other consortium members. This architectural advantage makes blockchain particularly well-suited for public procurement supervision, where the regulator itself may be subject to corruption pressures.

3.4. Robustness Check of Different Base Models

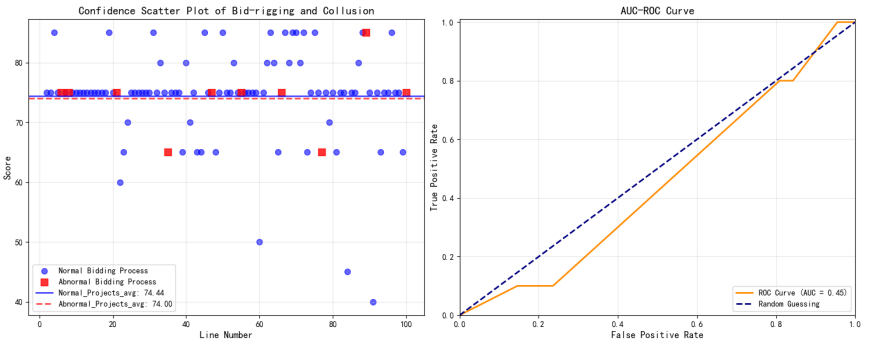
To validate model generalizability, we replicated the Zero-shot experiment using GPT-4o-mini. Results confirm robust detection capabilities: Figure 4(a) shows effective differentiation between abnormal and normal projects, consistent with GLM-4-Flash. While GPT-4o-mini demonstrates superior inference speed (avg. 7s vs. 30s, Figure 4d), GLM-4-Flash achieves slightly higher comprehensive accuracy (AUC and F1 scores, Figures 4b-c).Additionally, to assess the stability of the Few-shot strategy, we conducted sensitivity analyses by varying the exemplar sets across different random seeds (Seed = 42, ...). The results demonstrate that performance metrics fluctuated within a minimal range (F1-Score standard deviation < 0.02), confirming that the frameworks performance is not contingent upon the specific choice of demonstration examples. Overall, the frameworks efficacy and its dependency on soft information-holds across different foundation models and experimental configurations..

3.5. Time Cost and Efficiency Analysis

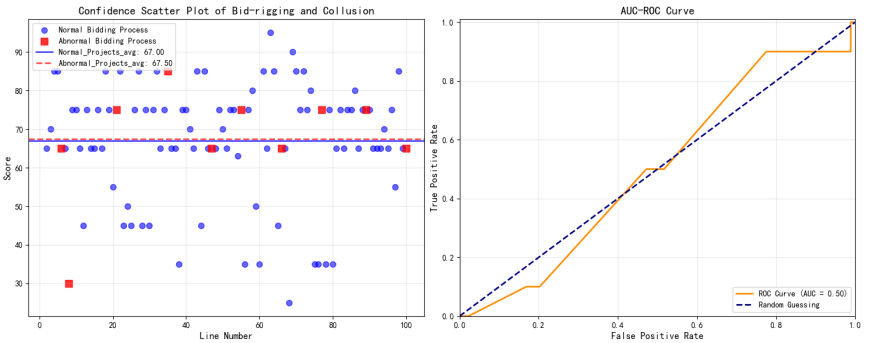
Finally, we evaluate the computational efficiency. Experimental results show that the Zero-shot method exhibits the lowest average inference time and the smallest variance. In contrast, the Chain-of-Thought method results in the highest time cost without bringing significant performance improvements (and still underperforming Logit in hard info tasks). From a cost-benefit perspective, the Zero-shot LLM architecture offers the optimal choice for deploying



(a) Zero-shot



(b) Few-shot



(c) Chain-of-Thought

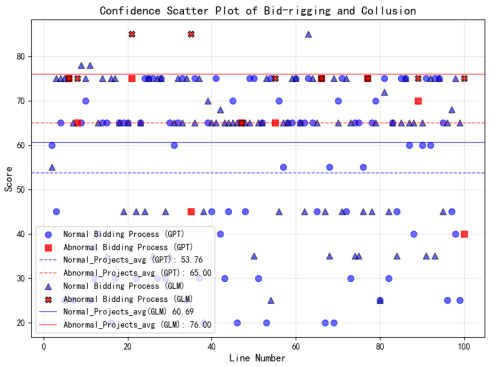
Figure 3: Performance Decay of LLMs Detection on Tampered Data Lacking Blockchain Evidence Storage

high-frequency, real-time RegTech monitoring systems.

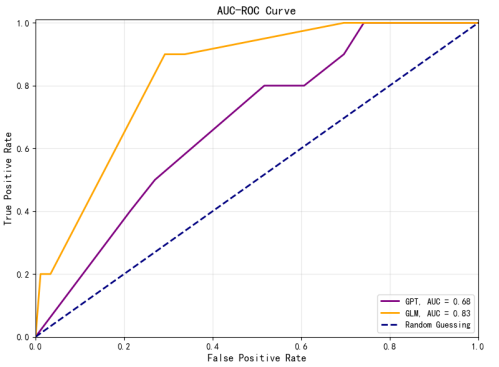
4. Mechanism Analysis and Discussion

4.1. Soft Information and Multi-dimensional Feature Extraction Mechanism

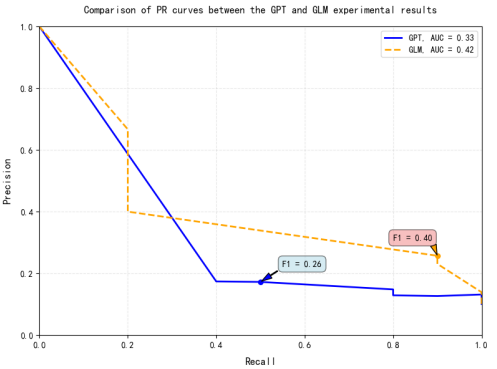
The frameworks superiority stems from the LLMs capacity to process "Soft Information. Unlike econometric models limited to numerical variance, LLMs function as multi-modal processors, mapping high-dimensional text features (e.g., semantic isomorphism in bid documents) into latent risk signals (Gentzkow, Kelly and Taddy, 2019). Critically, this capability transcends traditional static NLP methods such as TF-IDF or keyword matching, which operate on bag-of-words representations and can only detect surface-level lexical similarities. In contrast, LLMs perform cross-modal contextual reasoning, enabling them to identify deeper logical inconsistencies even when colluding bidders delib-



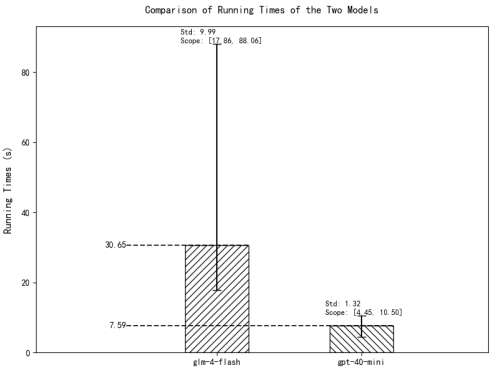
(a) Scatter Plot of Bid Rigging Confidence Under Different Models



(b) Comparison of AUC-ROC Curves Under Different Models



(c) Comparison of PR Curves Under Different Models



(d) Comparison of Running Time Under Different Models (Zero-shot Method)

Figure 4: Comparison of Detection Performance Under Different Base Models

erately vary technical terminology or phrasing. For instance, two bid documents may describe ostensibly different implementation approaches using distinct vocabulary, yet reveal identical underlying logic structuresa pattern invisible to frequency-based text analysis but detectable through LLMs semantic understanding. This reduces information asymmetry by capturing non-numerical anomaliessuch as duplicated contact details and contextual contradictionsthat traditional screenings miss.

4.2. The Occams Razor Effect: Why is the Zero-shot Strategy Superior?

A counter-intuitive finding in this study is that the simple Zero-shot strategy performs better than the complex Chain-of-Thought (CoT) and Few-shot strategies. This reflects the trade-off between cognitive load and signal-to-noise ratio in bidding supervision tasks. In bid rigging detection tasks, core criteria (e.g., extremely low price variance, repetition of key information) are often explicit and rule-oriented.

Although examples provided in Few-shot prompting increase context, they also introduce specific distributional biases. If the feature distribution of the sample to be tested is not completely consistent with the examples, the model may overfit to non-key details in the examples rather than focusing on general collusion logic. The Chain-of-Thought (CoT) strategy mandates the model to perform step-by-step reasoning. However, for tasks where logic is relatively intuitive, such as numerical comparison and keyword matching, mandatory long-chain reasoning may conversely cause the model to generate hallucinations or lose focus during intermediate steps.

4.3. ~~Technological Complementarity: Blockchain as a Regulatory Commitment Mechanism~~Error Analysis and Feature Attribution

~~The counterfactual experiment (Section 3.3) reveals a profound complementary relationship, rather than a substitution relationship, between Blockchain and Artificial Intelligence. Under traditional centralized storage architectures, regulators face the classic agency problem of who monitors the monitors. Internal personnel with database privileges can implement ex-post tampering, rendering LLM audits based on historical data ineffective. Therefore, the immutable nature of Blockchain acts as a critical institutional anchor. As emphasized in recent frameworks for secure e-government systems, such blockchain-based architectures are essential for guaranteeing data integrity and privacy against internal or external threats.~~

4.4. Limitations and Future Work

While this study demonstrates the potential of integrating Blockchain and LLMs for bid rigging detection, several limitations warrant acknowledgment and suggest directions for future research.

Model Constraints. Our empirical analysis relies on lightweight LLMs (GLM-4-Flash and GPT-4o-mini) due to computational cost considerations and API accessibility. These models, while efficient, may not fully exploit the semantic reasoning capabilities of more advanced architectures. Future research could explore the performance of larger-scale models (e.g., GPT-4, Claude-3) or emerging multi-modal large language models capable of simultaneously processing textual, numerical, and even visual information (e.g., scanned bid document images). Such models may achieve superior detection accuracy by capturing richer cross-modal patterns that lightweight models overlook.

Complementary Rather Than Substitutive Framework. We emphasize that our findings should not be interpreted as advocating for LLMs to entirely replace traditional econometric or machine learning models.Our contribution lies in proposing a framework where: (1) traditional statistical models serve as the first-line screen for hard information anomalies, and (2) LLMs function as a second-layer filter to capture soft information risks that escape numerical detection. This hybrid architecture balances detection efficacy with operational feasibility in real-world regulatory settings.

Expanded Scope of Collusion Detection. The current study focuses primarily on behavioral patterns observable within bid documents and pricing data. However, sophisticated bid rigging schemes often involve deeper corporate relationships that extend beyond individual procurement projects. Future research could integrate additional data sources, including: (1) equity ownership tracesdetecting hidden affiliations through shared shareholders, cross-holdings, or common ultimate beneficial owners; (2)capital flow analysisidentifying suspicious fund transfers between ostensibly independent bidders before or after contract awards; and (3)personnel mobility networkstracking executive or key employee movements across bidding entities. Incorporating such multi-layered relational data into the LLM framework could substantially enhance the detection of deeply concealed collusion networks.

5. Conclusion

Aiming at the chronic ailment of bid rigging that has long existed in the bidding process, this paper proposes a novel digital regulatory framework integrating Blockchain technology and Large Language Models (LLMs), and empirically investigates the effectiveness of this framework in identifying bid rigging behaviors. Research results confirm that while LLMs may underperform traditional statistical models (like Logit) on pure numerical tasks, they effectively complement this deficit by processing unstructured "soft information" in bid documents. By quantifying textual risks, the framework overcomes the theoretical ceiling of single-source screening, significantly improving the overall identification efficiency of bid rigging behaviors.

This study reaches two findings with important theoretical significance. First, regarding the choice of prompt engineering strategies, the simple Zero-shot strategy exhibits robustness superior to Few-shot and Chain-of-Thought (CoT) strategies in bidding supervision scenarios. Empirical data shows that overly long context prompts introduce additional noise (the Signal-to-Noise Ratio problem), causing models to suffer from attention dispersion when processing high-dimensional features. This indicates that in compliance audit tasks, model efficacy follows the "Less is More" principle. Second, the results of the counterfactual experiment emphasize the foundational status of Blockchain technology in algorithmic regulation. When underlying data suffers from ex-post tampering by centralized database administrators, even the most advanced LLMs degenerate into random guessing (with AUC dropping to approximately 0.5). Therefore, Blockchain is not merely a carrier for data storage but a necessary "Trust Anchor" ensuring the effectiveness of algorithmic governance; it solves the data credibility problem by locking the ex-ante information set.

Based on the above findings, this paper proposes the following policy recommendations: First, construct a dual regulatory system of "On-chain Evidence Storage + AI Audit." It is recommended that regulatory agencies require full-process data of public procurement projects to be put on the chain, utilizing the immutability of Blockchain to curb ex-post corruption, while simultaneously deploying lightweight LLMs to conduct real-time ex-ante early warning.Second, adopt a low-cost, high-frequency agile regulatory strategy. Given that the Zero-shot strategy significantly outperforms complex reasoning chains in both accuracy and speed, enterprises and governments do not need to invest huge capital in complex prompt engineering; utilizing general lightweight models directly can achieve efficient preliminary screening, thereby significantly enhancing audit efficacy and coverage rate without imposing heavy computational burdens.

Although this study verifies the framework's feasibility, two limitations persist. First, the dataset focuses on a specific sector in a southern city; thus, cross-regional and cross-industry generalizability requires further validation. Second, the absence of an industry-specific compliance lexicon limits the detection precision for "implicit collusion" (e.g., subtle textual cues). Future research should address these by: acquiring national-level multi-regional data to verify broad applicability; constructing a procurement compliance knowledge base to optimize prompt engineering; and developing ensemble models combining traditional Logit benchmarks with LLMs to maximize detection accuracy.

Data availability

The data analyzed in this study were obtained from a specific government procurement platform and involve sensitive commercial information. Due to confidentiality agreements and data privacy protection requirements, the dataset cannot be made publicly available.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Cao, S., Gunasekaran, A., Dubey, R., 2023. Blockchain technology in operations and supply chain management: A comprehensive overview. International Journal of Production Economics 263, 108960.

Chassang, S., Kawai, K., Nakabayashi, J., Ortner, J., 2022. Data-driven regulation: Theory and application to missing bids. Journal of Political Economy 130, 3156–3195.

Chod, J., Trichakis, N., Tsoukalas, G., et al., 2020. On the financing benefits of supply chain transparency and blockchain adoption. Information Systems Research 31, 1046–1064.

Cong, L.W., He, Z., 2019. Blockchain disruption and smart contracts. The Review of Financial Studies 32, 1754–1797.

Dowling, M., Lucey, B., 2023. Chatgpt for (finance) research: The bananarama conjecture. Finance Research Letters 53, 103662.

Elisa, N., Yang, L., Chao, F., et al., 2023. A framework of blockchain-based secure and privacy-preserving e-government system. Wireless Networks 29, 1005–1015.

Gentzkow, M., Kelly, B., Taddy, M., 2019. Text as data. Journal of Economic Literature 57, 535–574.

Goldstein, I., Spatt, C.S., Ye, M., 2021. Big data in finance. The Review of Financial Studies 34, 3213–3225.

Hansen, T., Kazinnik, R., 2023. Can chatgpt decipher fedspeak? Finance Research Letters 55, 103945.

Hu, H., Li, J., Liu, Y., 2022. The role of soft information in credit rating: Evidence from textual analysis. Finance Research Letters 47, 102661.

Huber, M., Imhof, D., Ishii, R., 2022. Transnational machine learning with screens for flagging bid-rigging cartels. Journal of the Royal Statistical Society Series A: Statistics in Society 185, 1074–1114.

Imhof, D., Karagök, Y., Rutz, S., 2018. Screening for bid riggingdoes it work? Journal of Competition Law & Economics 14, 235–261.

Kawai, K., Nakabayashi, J., 2022. Detecting large-scale collusion in procurement auctions. American Economic Review 112, 2193–2224.

Li, J., Pan, Y., 2023. Digital finance and corporate fraud: Evidence from china. Pacific-Basin Finance Journal 77, 101910.

Liberti, J.M., Petersen, M.A., 2019. Information: Hard and soft. The Review of Corporate Finance Studies 8, 1–41.

Luo, S., Li, J., 2023. Digital transformation and corporate fraud: Evidence from china. Finance Research Letters 52, 103518.

Nasir, M.A., Shaukat, K., Hameed, I.A., 2021. A bibliometric analysis of regtech: Current trends and future directions. Finance Research Letters 38, 101739.

Niszczota, P., Abbas, S., 2023. Gpt-4 as a financial analyst. Finance Research Letters 55, 104033.

Shen, H., Zhang, H., 2023. Machine learning in financial fraud detection: A comparative analysis. Finance Research Letters 55, 103900.

Signor, R., Ballesteros-Pérez, P., Love, P.E.D., 2023. Collusion detection in infrastructure procurement: A modified order statistic method for uncapped auctions. IEEE Transactions on Engineering Management 70, 464–477.

Signor, R., Love, P.E.D., Ballesteros-Pérez, P., 2024. Detecting bid rigging in public auctions for procuring infrastructure projects: Formulating the reference scenario for decision-making. Construction Management and Economics 42, 545–563.

Wu, K., Fu, Y., Kong, D., 2022. Does the digital transformation of enterprises affect stock price crash risk? Finance Research Letters 48, 102888.

Yue, P., Korkmaz, A.G., Zhou, H., 2023. Digital finance and corporate fraud: Evidence from china. Finance Research Letters 58, 104448.

~~Theoretical Derivation of "Blockchain+LLMs" Identification Efficiency~~

~~Definitions~~

~~Let A and B denote the binary events where a specific rigging behavior is successfully identified by the traditional Logit model (Hard Information) and the LLM (Soft Information), respectively. Let $P(A)$ and $P(B)$ represent their respective detection probabilities (Precision). The variances of these Bernoulli trials are defined as:~~

$$\mathrm{Var}(A) = P(A)(1 - P(A)) \quad \text{and} \quad \mathrm{Var}(B) = P(B)(1 - P(B))$$

~~Union Probability with Correlation~~

~~The total identification rate R of the "Blockchain+LLMs" mechanism is equivalent to the probability of the union of events A and B:~~

$$R = P(A \cup B) = P(A) + P(B) - P(A \cap B) \tag{2}$$

~~Unlike independent events, the detection capabilities of hard and soft information are correlated (e.g., price anomalies often coincide with text plagiarism). We introduce the correlation coefficient ρ to quantify this dependency. According to the definition of covariance for Bernoulli variables, the probability of the intersection $P(A \cap B)$ is expressed as:~~

$$P(A \cap B) = P(A)P(B) + \mathrm{Cov}(A, B)$$

Substituting the correlation formula $\mathrm{Cov}(A, B) = \rho\sqrt{\mathrm{Var}(A)\mathrm{Var}(B)}$:

$$P(A \cap B) = P(A)P(B) + \rho\sqrt{P(A)(1 - P(A)) \cdot P(B)(1 - P(B))} \tag{3}$$

~~Final Criterion and Complementarity Analysis~~

~~By substituting Eq (3) into Eq (2), the comprehensive identification efficiency function is derived as follows:~~

$$R = P(A) + P(B) - \left[P(A)P(B) + \rho\sqrt{P(A)(1 - P(A)) \cdot P(B)(1 - P(B))}\right] \tag{4}$$

~~From Eq (4), it is evident that the total identification rate R is strictly decreasing with respect to the correlation coefficient ρ:~~

$$\frac{\partial R}{\partial \rho} = -\sqrt{P(A)(1 - P(A)) \cdot P(B)(1 - P(B))} < 0$$

~~This derivative implies a fundamental principle of information aggregation in RegTech:~~

- ~~**High Correlation Scenario ($\rho \to 1$):** If price anomalies and text risks are perfectly collinear (i.e., riggers always expose both signals simultaneously), the intersection term is maximized. In this case, the LLM provides redundant information, offering minimal marginal gain.~~
- ~~**Low Correlation Scenario ($\rho \to 0$):** If hard and soft information are orthogonal (complementary), the redundancy is minimized. The system captures the maximum union of risk signals.~~

~~Therefore, the theoretical superiority of the "Blockchain + LLMs" framework relies on the *non-perfect correlation* between numerical and textual signals. It enables the system to detect sophisticated schemes where colluders successfully mask price anomalies (evading A) but leave textual traces (captured by B), thereby breaking the detection ceiling of single-source screening.~~